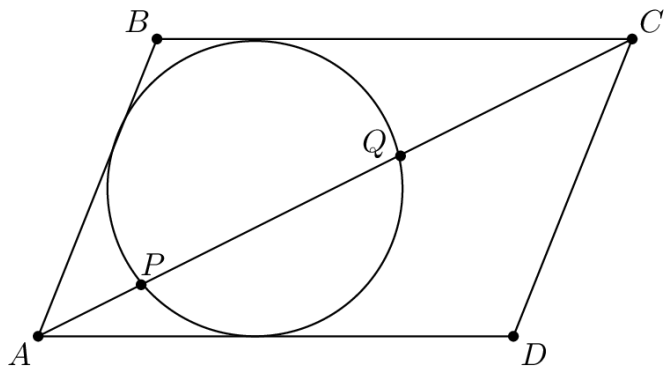


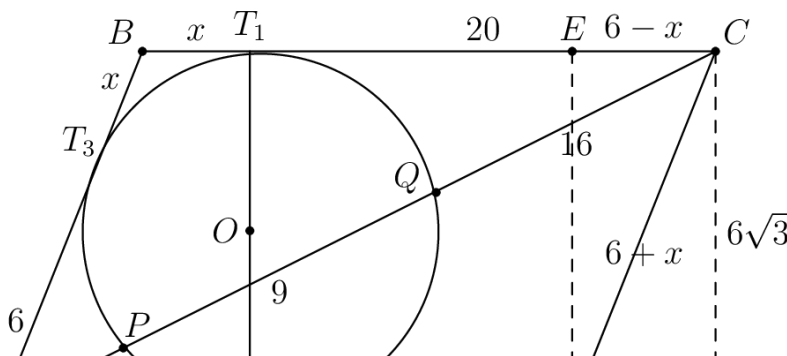
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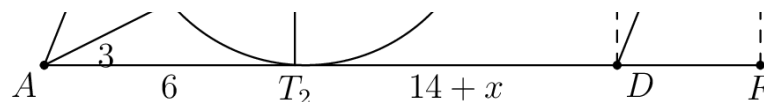
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Let $ABCD$ be a parallelogram with $\angle BAD < 90^\circ$. A circle tangent to sides $\overline{DA}$, $\overline{AB}$, and $\overline{BC}$ intersects diagonal $\overline{AC}$ at points P and Q with $AP < AQ$, as shown. Suppose that $AP = 3$, $PQ = 9$, and $QC = 16$. Then the area of $ABCD$ can be expressed in the form $m\sqrt{n}$, where m and n are positive integers, and n is not divisible by the square of any prime. Find $m + n$.



Let's redraw the diagram, but extend some helpful lines.





We obviously see that we must use power of a point since they've given us lengths in a circle and there are intersection points. Let T_1, T_2, T_3 be our tangents from the circle to the parallelogram. By the secant power of a point, the power of $A = 3 \cdot (3 + 9) = 36$. Then $AT_2 = AT_3 = \sqrt{36} = 6$. Similarly, the power of $C = 16 \cdot (16 + 9) = 400$ and $CT_1 = \sqrt{400} = 20$. We let $BT_3 = BT_1 = x$ and label the diagram accordingly.

Notice that because $BC = AD$, $20 + x = 6 + DT_2 \implies DT_2 = 14 + x$. Let O be the center of the circle. Since OT_1 and OT_2 intersect BC and AD , respectively, at right angles, we have T_2T_1CD is a right-angled trapezoid and more importantly, the diameter of the circle is the height of the triangle. Therefore, we can drop an altitude from D to BC and C to AD , and both are equal to $2r$. Since $T_1E = T_2D$, $20 - CE = 14 + x \implies CE = 6 - x$. Since $CE = DF$, $DF = 6 - x$ and $AF = 6 + 14 + x + 6 - x = 26$. We can now use Pythagorean theorem on $\triangle ACF$; we have $26^2 + (2r)^2 = (3 + 9 + 16)^2 \implies 4r^2 = 784 - 676 \implies 4r^2 = 108 \implies 2r = 6\sqrt{3}$ and $r^2 = 27$.

We know that $CD = 6 + x$ because $ABCD$ is a parallelogram. Using Pythagorean theorem on $\triangle CDE$,

$$(6+x)^2 = (6-x)^2 + 108 \implies (6+x)^2 - (6-x)^2 = 108 \implies 12 \cdot 2x = 108 \implies 2x = 9 \implies x = \frac{9}{2}$$

. Therefore, base $BC = 20 + \frac{9}{2} = \frac{49}{2}$. Thus the area of the parallelogram is the base times the height, which is

$$\frac{49}{2} \cdot 6\sqrt{3} = 147\sqrt{3}$$
 and the answer is 150

~KingRavi

Solution 2

Let the circle tangent to BC, AD, AB at P, Q, M separately, denote that $\angle ABC = \angle D = \alpha$

Using POP, it is very clear that $PC = 20$, $AQ = AM = 6$, let $BM = BP = x$, $QD = 14 + x$, using LOC in $\triangle ABP$, $x^2 + (x + 6)^2 - 2x(x + 6) \cos \alpha = 36 + PQ^2$, similarly, use LOC in $\triangle DQC$, getting that $(14 + x)^2 + (6 + x)^2 - 2(6 + x)(14 + x) \cos \alpha = 400 + PQ^2$. We use the second equation to minus the first equation, getting that $28x + 196 - (2x + 12) \times 14 \times \cos \alpha = 364$, we can get $\cos \alpha = \frac{2x - 12}{2x + 12}$.

Now applying LOC in $\triangle ADC$, getting $(6+x)^2 + (20+x)^2 - 2(6+x) \times (20+x) \times \frac{2x-12}{2x+12} = (3+9+16)^2$,
solving this equation to get $x = \frac{9}{2}$, then $\cos \alpha = -\frac{1}{7}$, $\sin \alpha = \frac{4\sqrt{3}}{7}$, the area is $\frac{21}{2} \cdot \frac{49}{2} \cdot \frac{4\sqrt{3}}{7} = 147\sqrt{3}$ leads to 150

~bluesoul,HarveyZhang

Solution 3

Denote by O the center of the circle. Denote by r the radius of the circle. Denote by E, F, G the points that the circle meets AB, CD, AD at, respectively.

Because the circle is tangent to $AD, CB, AB, OE = OF = OG = r, OE \perp AD, OF \perp CB, OG \perp AB$.

Because $AD \parallel CB$, E, O, F are collinear.

Following from the power of a point, $AG^2 = AE^2 = AP \cdot AQ$. Hence, $AG = AE = 6$.

Following from the power of a point, $CF^2 = CQ \cdot CP$. Hence, $CF = 20$.

Denote $BG = x$. Because DG and DF are tangents to the circle, $BF = x$.

Because $AEFB$ is a right trapezoid, $AB^2 = EF^2 + (AE - BF)^2$. Hence, $(6 + x)^2 = 4r^2 + (6 - x)^2$. This can be simplified as

$$6x = r^2. \quad (1)$$

In $\triangle ACB$, by applying the law of cosines, we have

$$\begin{aligned} AC^2 &= AB^2 + CB^2 - 2AB \cdot CB \cos B \\ &= AB^2 + CB^2 + 2AB \cdot CB \cos A \\ &= AB^2 + CB^2 + 2AB \cdot CB \cdot \frac{AE - BF}{AB} \\ &= AB^2 + CB^2 + 2CB(AE - BF) \\ &= (6 + x)^2 + (20 + x)^2 + 2(20 + x)(6 - x) \\ &= 24x + 676. \end{aligned}$$

Because $AC = AP + PQ + QC = 28$, we get $x = \frac{9}{2}$. Plugging this into Equation (1), we get $r = 3\sqrt{3}$.

Therefore,

$$\begin{aligned} \text{Area } ABCD &= CB \cdot EF \\ &= (20 + x) \cdot 2r \\ &= 147\sqrt{3}. \end{aligned}$$

Therefore, the answer is $147 + 3 = \boxed{(150)}$.

~Steven Chen (www.professorchenedu.com)

Solution 4

Let ω be the circle, let r be the radius of ω , and let the points at which ω is tangent to AB , BC , and AD be X , Y , and Z , respectively. Note that PoP on A and C with respect to ω yields $AX = 6$ and $CY = 20$. We can compute the area of ABC in two ways:

1. By the half-base-height formula, $[ABC] = r(20 + BX)$.

2. We can drop altitudes from the center O of ω to AB , BC , and AC , which have lengths r , r , and $\sqrt{r^2 - \frac{81}{4}}$. Thus,

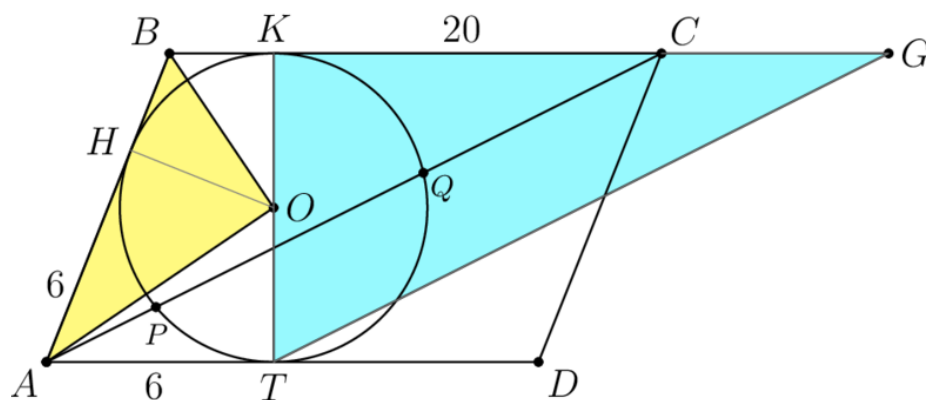
$$[ABC] = [OAB] + [OBC] + [OAC] = r(BX + 13) + 14\sqrt{r^2 - \frac{81}{4}}.$$

Equating the two expressions for $[ABC]$ and solving for r yields $r = 3\sqrt{3}$.

Let $BX = BY = a$. By the Parallelogram Law, $(a + 6)^2 + (a + 20)^2 = 38^2$. Solving for a yields $a = 9/2$. Thus, $[ABCD] = 2[ABC] = 2r(20 + a) = 147\sqrt{3}$, for a final answer of $\boxed{150}$.

~ Leo.Euler

Solution 5



Let ω be the circle, let r be the radius of ω , and let the points at which ω is tangent to AB , BC , and AD be H , K , and T , respectively. PoP on A and C with respect to ω yields

$$AT = 6, CK = 20.$$

Let $TG = AC, CG \parallel AT$.

$$\text{In } \triangle KGT \quad KT \perp BC, \quad KT = \sqrt{GT^2 - (KC + AT)^2} = 6\sqrt{3} = 2r.$$

$$\angle AOB = 90^\circ, OH \perp AB, OH = r = \frac{KT}{2},$$

$$OH^2 = AH \cdot BH \implies BH = \frac{9}{2}.$$

Area is

$$(BK + KC) \cdot KT = (BH + KC) \cdot 2r = \frac{49}{2} \cdot 6\sqrt{3} = 147\sqrt{3} \implies 147 + 3 = \boxed{150}.$$

vladimir.shelomovskii@gmail.com, vvsss

Solution 6 (Short and Sweet)

Let O be the center of the circle. Let points M , N and L be the tangent points of lines BC , AD and AB respectively to the circle. By Power of a Point, $(MC)^2 = 16 \cdot 25 \implies MC = 20$. Similarly, $(AL)^2 = 3 \cdot 12 \implies AL = 6$. Notice that $AL = AN = 6$ since quadrilateral $LONA$ is symmetrical. Let AC intersect MN at I . Then, $\triangle IMC$ is similar to $\triangle AIN$.

Therefore, $\frac{CI}{MC} = \frac{AI}{AN}$. Let the length of $PI = l$, then $\frac{25-l}{20} = \frac{3+l}{6}$. Solving we get $l = \frac{45}{13}$. Doing the Pythagorean theorem on triangles IMC and AIN for sides MI and IN respectively, we obtain the equation

$$\sqrt{\left(\frac{280}{13}\right)^2 - 400} + \sqrt{\left(\frac{84}{13}\right)^2 - 36} = MN = 2r_1 \text{ where } r_1 \text{ denotes the radius of the circle. Solving, we get } MN = 6\sqrt{3}.$$

Additionally, quadrilateral $OLBM$ is symmetrical so $OL = OM$. Let $OL = OM = x$ and extend a perpendicular foot from B to AD and call it R . Then, $\triangle ABR$ is right with $AR = 6 - x$, $AB = 6 + x$, and $RB = 2r_1 = MN = 6\sqrt{3}$. Taking the difference of squares, we get $108 = 24x \implies x = \frac{9}{2}$. The area of $ABCD$ is

$$MN \cdot BC = (20 + x) \cdot MN \implies \frac{49}{2} \cdot 6\sqrt{3} = 147\sqrt{3}. \text{ Therefore, the answer is } 147 + 3 = \boxed{150}$$

~Magnetoininja

Solution 7 (Intuitive, no trig, no weird auxiliary lines)

Say that BC is tangent to the circle at X and AD tangent at Y . Also, H is the intersection of XY (diameter) and AC (diagonal). Then

by power of a point with given info on A and C we get that $AY = 6$ and $CX = 20$. Note that $HAY \sim HCX$, and since

$$\frac{AY}{CX} = \frac{3}{10} \text{ we note that}$$

$$\frac{AH}{CH} = \frac{AP + PH}{CQ + QH} = \frac{3 + PH}{16 + QH} = \frac{AY}{CX} = \frac{3}{10}$$

. Since $PH + HQ = 9$, we get that $PH = \frac{45}{13}$ and $QH = \frac{72}{13}$. This is the length information within the circle. The same triangle

similarity also means that $\frac{YH}{XH} = \frac{3}{10}$, so if the radius of the circle is r then we have $XH = \frac{20}{13}r$ and $YH = \frac{6}{13}r$. By power of a point on H, we can figure out r :

$$XH \cdot YH = PH \cdot QH$$

$$\frac{20}{13}r \cdot \frac{6}{13}r = \frac{45}{13} \cdot \frac{72}{13}$$

and we get that $r = 3\sqrt{3}$. Thus, we have that the height of the parallelogram is $2r = 6\sqrt{3}$ and we want to find BC . If AB is tangent to the circle at E , then set $a = BX = BE$. Using pythagorean theorem, $AO^2 + BO^2 = AB^2$ and we can plug in diagram values:

$$(AY^2 + OY^2) + (BX^2 + OX)^2 = AB^2$$

$$(6^2 + (3\sqrt{3})^2) + (a^2 + (3\sqrt{3})^2) = (a + 6)^2.$$

$$\text{Solving, we get } a = \frac{9}{2} \text{ Finally, we have } [ABCD] = XY \cdot BC = 6\sqrt{3} \cdot (20 + \frac{9}{2}) \rightarrow \boxed{150}$$

~ Brocolimanx

Solution 8 (Ptolemy's Theorem + Power of Point + Pythagorean Theorem)

Let E, F, G be the circle's point of tangency with sides AD, AB , and BC , respectively. Let O be the center of the inscribed circle.

By Power of a Point, $AE^2 = AP \cdot AQ = 3(3 + 9) = 36$, so $AE = 6$. Similarly,

$GC^2 = CQ \cdot CP = 16(16 + 9) = 400$, so $GC = 20$.

Construct GE , and let I be the point of intersection of GE and AC . $GE \perp BC$ and $GE \perp AD$. By AA, $\triangle IGC \sim \triangle IEA$, and we have $\frac{AI}{IC} = \frac{AE}{GC} = \frac{3}{10}$. We also know $AI + IC = AC = 28$, so $AI = \frac{84}{13}$ and $IC = \frac{280}{13}$.

Using Pythagorean Theorem on $\triangle IEA$ and $\triangle CIG$, we find that $EI = \frac{18\sqrt{3}}{13}$ and $IG = \frac{60\sqrt{3}}{13}$. Thus,

$GE = EI + IG = 6\sqrt{3}$, and the radius of the circle is $3\sqrt{3}$.

Construct EF, FG . $\angle AFO = \angle AEO = 90^\circ$, so $AEOF$ is cyclic. Similarly, $BFOG$ is cyclic.

Now, we attempt to set up Ptolemy. Using Pythagorean Theorem on $\triangle AEO$, we find that $AO = 3\sqrt{7}$. By Ptolemy's Theorem, $(AE)(FO) + (AF)(EO) = (AO)(FE)$, from which we have $(6)(3\sqrt{3}) + (6)(3\sqrt{3}) = (3\sqrt{7})(FE)$ and

$FE = 12\frac{\sqrt{3}}{\sqrt{7}}$. From Thales' Circle, $\triangle FGE$ is a right triangle, and $EF^2 + FG^2 = GE^2$, so $FG = \frac{18}{\sqrt{7}}$.

Set $BF = BG = s$. $BO = \sqrt{s^2 + (3\sqrt{3})^2} = \sqrt{s^2 + 27}$, so by Ptolemy's Theorem on $BFOG$, we have

$$(BF)(GO) + (BG)(FO) = (FG)(BO)$$

$$(3\sqrt{3})(s) + (3\sqrt{3})(s) = (\frac{18}{\sqrt{7}})(\sqrt{s^2 + 27})$$

Solving yields $s = \frac{9}{2}$.

We know that $BC = BG + GC = 20 + \frac{9}{2} = \frac{49}{2}$, so the area of $ABCD = (\frac{49}{2})(6\sqrt{3}) = 147\sqrt{3}$. The requested answer is $147 + 3 = \boxed{150}$.

~ adam_zheng

Solution 9

Let points E, F, G be the points where lines BC, AB , and AD are tangent to the circle respectively. Then extend line BC to point H such that AH is perpendicular to BC .

According to Power of a Point, $CQ \cdot CP = CE^2 \rightarrow 16 \cdot 25 = CE^2 \rightarrow CE = 20$.

Similarly, $AP \cdot AQ = AF^2 \rightarrow 3 \cdot 12 = AF^2 \rightarrow AF = 6$.

(Note that $AG = AF = 6$ because they are intersecting tangents from the same circle. Furthermore, $HE = AG = 6$ due to a simple upwards translation.)

Therefore, $HC = HE + EC = 6 + 20 = 26$, and we already know that $AC = 28$, meaning that we can use the Pythagorean Theorem on $\triangle AHC$ to obtain: $AH = 6\sqrt{3}$.

Let $HB = k$. Then $BE = 6 - HB = 6 - k$. Since they are intersecting tangents from the same circle, $BF = BE = 6 - k$.

Therefore, $AB = AF + BF = 6 + 6 - k = 12 - k$. We have all the side lengths of $\triangle ABH$, so applying the Pythagorean Theorem: $(6\sqrt{3})^2 + k^2 = (12 - k)^2 \rightarrow k = \frac{3}{2}$.

Thus, $BC = BE + EC = 6 - k + 20 = 26 - k = \frac{49}{2}$.

$[ABCD] = AH \cdot BC = 6\sqrt{3} \cdot \frac{49}{2} = 147\sqrt{3}$, so the answer is $\boxed{150}$.

~sid2012

Video Solution

https://www.youtube.com/watch?v=FeM_xXiJ0c&t=1s

~Steven Chen (www.professorchenedu.com)

Video Solution 2 (Mathematical Dexterity)

<https://www.youtube.com/watch?v=1nDKQkr9NaU>

Video Solution 3 by OmegaLearn

<https://youtu.be/LpOegT0fKy8?t=740>

~ pi_is_3.14

Video Solution 4 by StressedPineapple

<https://youtube.com/watch?v=x66z0IMj0as&t=640s>

Video Solution 5 by Continuum Math (PoP + Pythag ONLY)

<https://youtu.be/i5JA9ossq3o?si=IUZN4r7tg6mIC6cj>

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